

context_compactor.sh – Companion Guide

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Overview

§11.4.141 M7 implementation. Prunes stale tool-results and old thinking from long sessions. NEVER touches the governance prefix.

Functions

Function	Purpose	Example
<code>cc_should_compact</code>	Check if context exceeds threshold	<code>cc_should_compact 50</code>
<code>cc_prune_stale_results</code>	Mark old results as prunable	<code>cc_prune_stale_results 20</code>
<code>cc_summarize_old_thinking</code>	Compress old thinking blocks	<code>cc_summarize_old_thinking</code>
<code>cc_preserve_governance</code>	Documented invariant (no-op)	<code>cc_preserve_governance</code>
<code>cc_get_context_stats</code>	Return context statistics	<code>cc_get_context_stats</code>

Usage

```
source constitution/scripts/token_efficiency/context_compactor.sh
```

```
# Check if compaction needed (after 50 turns)
```

```
if cc_should_compact 50; then
```

```
    cc_prune_stale_results 20
```

```
    cc_summarize_old_thinking
```

```
fi
```

```
# Governance preservation is automatic (invariant)
```

Invariant

The governance prefix (CLAUDE.md, Constitution.md) is NEVER touched by compaction. It is the cache breakpoint and must remain byte-stable for prompt caching.

Cross-references

- Constitution §11.4.141 M7
- `token_efficiency_lib.sh` — for status and caching
- `scoped_read.sh` — for targeted reads